Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

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Disclosures and Notes

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Notes on website version

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Acknowledgements

Preface

Outline of the dissertation

Papers in the dissertation

Study I

Thomsen, H. B., Lebiecka-Johansen, B., Nørgaard, O., Andersen, T. H., Andersen, S. T., Fagherazzi, G., Hulman, A., & Isaksen, A. A. (2026). Continuous Glucose Monitoring–Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review. JMIR Diabetes, 11(1), e89374. https://doi.org/10.2196/89374

Study II

Thomsen, H. B., Andersen, S. T., Fagherazzi, G., Isaksen, A. A. & Hulman, A. (2026). Foundation model–derived representations of continuous glucose monitoring data: relation to glycemic patterns and cardiovascular risk. (Under review at JMIR AI, (7.30.2026), pending for preprint at JMIR preprints)

Study III

Thomsen, H. B., Li, L. Y., Isaksen, A. A., Lebiecka-Johansen, B., Bour, C., Fagherazzi, G., Doorn, W. P. T. M. van, Varga, T. V., & Hulman, A. (2025). Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes. PLOS Digital Health, 4(6), e0000918. https://doi.org/10.1371/journal.pdig.0000918

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Abbreviations

Introduction 1

1.1 Diabetes

☐ Check the sources from ADA_2026_ch2 to see if you can elaborate on the race thing and cite the original articles and not the summary ADA gives

Diabetes is a metabolic disease characterized by high blood glucose levels known as hyperglycemia.[1] The disease is one of the fastest growing global healths challenges.[2] In 2021 it was estimated that 537 million people lived with diabetes, and it is expected to reach 643 million by 2030.[2] The most common forms of chronic diabetes are type 1 diabetes (T1D) and type 2 diabetes (T2D).[1] T1D is an autoimmune disease characterized by the destruction of the pancreatic islet β-cells, leading to an absolute insulin deficiency.[3] Approximately 85%-95% of all people with diabetes in developing countries have T2D.[1] It is the most common form of diabetes and in contrast to T1D it generally begins with insulin resistance in which the body's cells become less responsive to insulin.[4,**Sachs2023?**] Initially, the pancreas compensates by increasing insulin secretion, but over time progressive β-cell dysfunction results in inadequate insulin production to meet the body's demands.[1,4,5] T2D is the most which is is Despite their distinct underlying pathophysiology, both T1D and T2D are characterized by chronic hyperglycemia.

Diabetes can be diagnosed based on a bloodtest to check hemoglobin A1c (HbA1c) levels or through plasma glucose criteria (Table 1.1).[3,**ADA2023_ch6?**]

Although both plasma glucose tests and HbA1c tests are appropriate for diabetes screening, the HbA1c test has several advantages.[3] Some of the benefits are that it requires no fasting and it is less sensitive to stress and changes in nutrition.[3] HbA1c is an indirect measure of blood glucose levels because glucose binds hemoglobin over $\sim$ 120 days lifespan of erythrocytes (red blood cells).[3] However HbA1c testing is not always accessible, affordable or suitable (e.g., for people with anemia or under HIV-treatment).[3] With the guidelines recommendation of using HbA1c, a debate has sparked on the controversy of using a set thresholds as it is well known that racial differences in HbA1c exists.[6] Black individuals have higher HbA1c levels than Hispanic and non-Hispanic White people with diabetes,[3,7] however studies show that the association between HbA1c and risk of complications appear to be similar between ethnic race groups.[3]

1.2 Blood Glucose Monitoring

HbA1c values are also a very important part of diabetes management. The treatment goal is for HbA1c to be below 53 mmol/mol for non-pregnant adults without any hyperglycemia or hypoglycemia (low blood glucose levels). However HbA1c goals should be individualized to fit each individuals specific situation HbA1c can vary from e.g., <48mmol/mol to none.[8,9] There are two tools available for self-assessment of glycemia, the self-monitoring blood glucose (SMBG) device measuring capillary blood with finger-sticks, and the continuous glucose monitoring device (CGM) measuring interstitial fluid continuously.[9,10] Both monitoring devices have shown to

Table 1.1: Diagnostic tests for diabetes

Test	Criteria	Description
Fasting plasma glucose	≥7.0 [mmol/L]	Fasting is defined as no calorie intake for at least eight hours.
2-hour plasma glucose	≥11.1 [mmol/L]	Glucose is measured 2 hours after oral ingestion of 75g glucose dissolved in water (OGTT)
Random plasma glucose	≥11.1 [mmol/L]	Individuals with classic symptoms of hyperglycemia (e.g., dehydration and unintentional weight loss) or hyperglycemic crisis (e.g., diabetic ketoacidosis) together with a plasma test taken any time of day.
Glycated hemoglobin (HbA1c)	≥48 [mmol/mol]	

improve glycemic control in people with diabetes, however the CGM is superior when it comes to immediate information and alerts on asymptomatic hypo- and hyperglycemic events and provide a comprehensive glucose profile including both pre- and postprandial blood glucose measurements.[9,11,12]

1.3 Continuous Glucose Monitoring Devices

The CGM devices measures the glucose in the interstitial fluid, which is the fluid around and between cells. It corresponds with plasma glucose although a lag can appear when glucose levels rise or fall rapidly.[ADA2026_ch7?]

1.4 Management

1.5 Glycemic markers and glucose monitoring

- ☒ HbA1c and its limitations
- ☒ Continuous glucose monitoring (CGM)
- ☐ AIDs use CGMs and algorithms to adjust insulin delivery.[ADA2026_ch7?]
- [ ]Glycemic variability
  - ☐ Metrics of glycemic variability
  - ☐ Clinical challenges of glycemic variability (Think HbA1c and MBG and variability study)

1.6 Cardiovascular diseases and Diabetes

- Cardiovascular complications
- Heart failure in diabetes

Williams2020 expenditures and complications: https://www.sciencedirect.com/science/article/pii/S0168822720301388 ## Cardiovascular risk prediction in diabetes * Traditional risk models * SCORE2-Diabetes risk score * What is SCORE2-Diabetes 10 year CVD risk score? * Limitations of conventional predictors

1.7 Machine learning

Relevant? Artificial Intelligence: The Future for Diabetes Care: https://www.clinicalkey.com/#!/content/playContent/1-s2.0-S0002934320303399?returnurl=https:%2F%2Flinkinghub.elsevier.

- Data-driven vs hypothesis-drivenly(knowledge-based) paradigms
- Why machine learning in diabetes and cardiovascular risk prediction?
- Why are ML fantastic for CGM data?
- Types of machine learning methods
  - Logistic regression
  - LSTM and CNNs
  - Foundation models and their advantage
  - The CGMformer is bla bla bla[13]
- Challenges and pitfalls of machine learning in healthcare
  - Overfitting and generalizability
  - Interpretability and explainability

1.8 Fairness and bias in machine learning

- Define race and ethnicity
- Algorithmic bias and group fairness
- Fairness metrics and mitigation strategies
- Trends in the quality of care and racial disparities in Medicare managed care: https://pubmed.ncbi.nlm.nih.gov/16107622/

2 Specific Aims

Aims {#sec-aims}

The overall aim of this PhD dissertation was to investigate the potential of continuous glucose monitoring (CGM) data to improve cardiovascular risk assessment and to evaluate the role of machine learning in extracting clinically relevant information from CGM data across different diabetes populations.

Aim 1: To identify and summarize the existing literature on the associations between glycemic control and cardiovascular outcomes, including the use of CGM-derived metrics for cardiovascular risk prediction. *(Study I)*

Aim 2: To determine whether a CGM foundation model captures meaningful and interpretable glucose representations in individuals with type 1 and type 2 diabetes. *(Study II)*

Aim 3: To evaluate whether conventional CGM-derived metrics and learned glucose representations improve prediction of elevated NT-proBNP beyond established clinical risk factors in individuals with type 2 diabetes. *(Study II)*

Aim 4: To evaluate the fairness of machine learning models for glucose prediction and compare strategies for tailoring prediction models to individual characteristics. *(Study III)*

References

1. Sacks DB, Arnold M, Bakris GL, et al. Guidelines and recommendations for laboratory analysis in the diagnosis and management of diabetes mellitus. Diabetes care 2023;46(10):e151–99.

2. Magliano DJ, Boyko EJ. IDF diabetes atlas [Internet]. 10th ed. Brussels, Belgium: International Diabetes Federation; 2021. Available from: https://www.ncbi.nlm.nih.gov/books/NBK581930/

3. American Diabetes Association Professional Practice Committee for Diabetes*. 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes—2026. Diabetes Care [Internet] 2025 [cited 2026 July 8];49(Supplement_1):S27–49. Available from: https://doi.org/10.2337/dc26-S002

4. Hall ME, Hall JE. Guyton and hall textbook of medical physiology. Elsevier; 2020.

5. Deshpande AD, Harris-Hayes M, Schootman M. Epidemiology of diabetes and diabetes-related complications. Physical therapy 2008;88(11):1254–64.

6. Selvin E, Rawlings AM, Bergenstal RM, Coresh J, Brancati FL. No Racial Differences in the Association of Glycated Hemoglobin With Kidney Disease and Cardiovascular Outcomes. Diabetes Care [Internet] 2013 [cited 2026 July 9];36(10):2995–3001. Available from: https://doi.org/10.2337/dc12-2715

7. Kirk JK, D'Agostino RB, Bell RA, et al. Disparities in HbA1c Levels Between African-American and Non-Hispanic White Adults With Diabetes. Diabetes care [Internet] 2006 [cited 2026 July 9];29(9):2130–6. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC3557948/

8. American Diabetes Association Professional Practice Committee for Diabetes*. 6. Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises: Standards of Care in Diabetes—2026. Diabetes Care [Internet] 2025 [cited 2026 July 9];49(Supplement_1):S132–49. Available from: https://doi.org/10.2337/dc26-S006

9. International Diabetes Federation. IDF global clinical practice recommendations for managing type 2 diabetes [Internet]. Brussels, Belgium: International Diabetes Federation; 2025. Available from: https://idf.org/download-global-clinical-practice-recommendations/

10. Buscemi S, Re A, Batsis J A, et al. Glycaemic variability using continuous glucose monitoring and endothelial function in the metabolic syndrome and in Type 2 diabetes. Comment in: Diabet Med 2011 Jan;28(1):125-6; author reply 127-8 PMID: 21166857 [https://wwwncbinlmnihgov/pubmed/21166857] Comment in: Diabet Med 2011 Jan;28(1):126; author reply 127-8 PMID: 21166858 [https://wwwncbinlmnihgov/pubmed/21166858] [Internet] 2010;27(8):872–8. Available from: http://ovidsp.ovid.com/ovidweb.cgi?T=JS&PAGE=reference&D=med8&NEWS=N&AN=20653743

11. Danne T, Nimri R, Battelino T, et al. International consensus on use of continuous glucose monitoring. Diabetes care 2017;40(12):1631–40.

12. Freckmann G, Schauer S, Beltzer A, et al. Continuous Glucose Profiles in Healthy People With Fixed Meal Times and Under Everyday Life Conditions. Journal of Diabetes Science and Technology 2024;18(2):407–13.

13. Lu Y, Liu D, Liang Z, et al. A pretrained transformer model for decoding individual glucose dynamics from continuous glucose monitoring data. National science review 2025;12(5):nwaf039–.